Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building production **REST APIs** and services with JWT auth, caching, fallbacks, and failure isolation on Docker and **Kubernetes**, plus authentication security research, ramping quickly in **Go**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant production FastAPI **background services** with explicit job-state transitions and failure isolation, processing 1K+ AI jobs at 3x throughput and preventing cascading failures under concurrent load.
- Designed REST-backed integrations across PostgreSQL, SQL Server, and cloud storage with validation and error handling, holding 99%+ **reliability** over 5K+ monthly jobs and cutting dashboard latency 71%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Flask **REST APIs** with Blueprint-based modules for auth, uploads, and grading behind a React client, serving 10K+ daily requests for 150+ instructors with stable latency at peak load.
- Added **timeout-aware fallback** routing across GPT-4o, Claude, and Qwen-VL external model providers and per-call token and cost logging, plus concurrent batch ingestion of 1K+ files with fault-tolerant execution.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned large-scale **API** responses for low-bandwidth clients with pagination, page and limit parameters, field filtering, indexing, and caching, improving response times 30% and cutting technical debt 20%.
- Automated nightly Jenkins pipelines with 20+ unit and integration **test** suites across Windows, Linux, and macOS, collaborating with Microsoft and GitHub stakeholders across India and US teams.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight Node.js **microservices** with REST APIs, including JWT authentication and role-based access control services, synchronizing 2M+ records across distributed locations through real-time WebSocket updates with API caching.
- Shipped Docker builds through Jenkins **CI/CD** with Snyk and OWASP dependency scanning and rolling zero-downtime releases, monitoring errors, latency, and logs with Sentry, Azure Monitor, and Kibana.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Python Flask REST APIs for bookings, pricing, coupons, and payments, integrating existing AWS-hosted auth and geo-data microservices and **JWT** sessions, sustaining sub-200ms latency across 50K+ booking records.
- Implemented idempotent payment **webhooks** for refunds, disputes, and chargebacks plus passwordless wallet-signature authentication via web3.js, keeping payment status consistent across retries and cutting manual payment handling by 70%.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, Shell
APIs & Backend: REST API Design, HTTP Semantics, FastAPI, Flask, Node.js, Express.js, Microservices, Webhooks, WebSockets
Reliability: Retries and Fallbacks, Failure Isolation, Caching, Pagination, Latency Optimization, Monitoring, Incident Troubleshooting
Auth & Security: JWT, RBAC, OAuth 2.0, Row-Level Security, Snyk, OWASP Dependency-Check, Wallet-Signature Auth
Platform: Docker, Kubernetes, NGINX, AWS, Azure, CI/CD, Jenkins, GitHub Actions, Sentry, Kibana
Data: PostgreSQL, SQL Server, MongoDB, Firebase

Projects

Contract-First API Gateway | *Go, gRPC, OpenAPI, Redis, Kubernetes, Prometheus*

- Designed a Go API gateway that routes REST and gRPC traffic defined in OpenAPI and protobuf, enforcing JWT and API-key auth, Redis token-bucket **rate limiting**, retries with circuit breaking, and Prometheus metrics.

FinNexus Live | *Kubernetes, Next.js, TypeScript, MongoDB, WebSockets*

- Built a TypeScript and Next.js platform on **Kubernetes** with a token-based firewall gating transaction APIs, Plaid integration, and WebSocket updates, serving 50K+ users across 1M+ transactions via CI/CD.

Comprehensive Analysis of Blockchain for User Authentication Paper | *RBAC, Smart Contracts, Self-Sovereign Identity*

- Co-authored an IEEE paper on blockchain-based **authentication**, analyzing smart-contract verification, role-based access, IoT device identity, and self-sovereign identity against identity theft, replay, man-in-the-middle, and phishing attacks on users.